📊 Executive Summary: Business Performance & Growth Analysis

Prepared By: Nishit Nagpal | Data Scientist & BI Analyst
Project: Dynamic Business Performance Dashboard

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1. Project Objective (Why This Was Built)

The primary objective of this project was to transition the organisation from static, siloed Excel reporting into an automated, interactive Business Intelligence solution. By leveraging a robust Star Schema data model and advanced DAX time-intelligence measures, this dashboard was designed to allow executives to instantly track Year-to-Date (YTD) versus Prior Year-to-Date (PYTD) performance, identify high-margin accounts, and isolate underperforming regions.

2. Key Business Insights (What We Found)

Based on the dashboard's analysis of the `[2022-2024]` dataset, several key trends emerged regarding our sales volume and profitability:

- **Overall YTD vs PYTD Variance:** Total sales for the year 2024 reached `$1.40 Million`, representing a `negative` growth of `4.76%` compared to the previous year. However, Gross Profit only shifted by `0.47%`, indicating a relatively stable overall margin structure despite the drop in sales volume.
- **Segmenting High-Value Accounts (The Scatter Plot):** By analysing Gross Profit % against YTD Sales, we identified a cluster of "High-Volume, Low-Margin" accounts in the segment, namely `Grant, Murphy and O'Kon; Behan, Mohr and Hyatt, Marvin Group, and Runte and Sons, among other notable groups`. Conversely, `Watsica LLC, Ortiz LLC, Heaney, Schuppe and Ward, among the notable groups,` showed lower total sales but exceptionally high margins, presenting a massive opportunity for scalable growth.
- **Geographic Pain Points (The Tree Map):** The Bottom 10 Countries visualisation revealed that `Canada`, `Germany`, and `Japan` experienced the sharpest decline in PYTD variance, dropping by an average of `29%`.
- **Product Line Trends (The Waterfall Chart):** The YTD variance waterfall chart highlights that the `Landscape Products` family was the primary driver of positive growth, contributing `27k` to the overall increase, offsetting the losses seen in the `Indoor Products` category.

3. Strategic Takeaways (Actionable Recommendations)

To translate these data points into business action, the following strategies are recommended:

1. **Margin Optimisation:** We need to renegotiate pricing or reduce operational costs for the high-volume, low-margin accounts identified in the scatter plot to bring them above the benchmark average line.
2. **Geographic Pivot:** Marketing and sales resources should be temporarily reallocated from the bottom-performing regions (like `[Country 1]`) to double down on regions showing high Gross Profit % growth.
3. **Inventory Alignment:** Production should be scaled up for the `[Top Product Family]` to meet the clear upward trend in Quantity YTD, ensuring no stockouts during peak quarters.

4. Forecasting & Future Outlook

With this foundational historical data modelled properly, the business is now positioned to move from descriptive analytics (what happened) to predictive analytics (what will happen).

- **Next Steps:** The next iteration of this reporting suite should involve exporting this cleaned dataset into Python/Scikit-Learn to build a time-series forecasting model (e.g., ARIMA or Prophet). This will allow us to predict Q4 sales volume and optimise supply chain logistics ahead of time.

5. Developer Retrospective (What I Learned)

From a technical engineering standpoint, this project reinforced several advanced BI concepts:

- **Complex State Management via DAX:** Implementing dynamic `SWITCH` measures allowed me to consolidate what would normally be 15 different charts into 5 dynamic visuals. This drastically improved the dashboard's load time and user experience.
- **Data Modelling:** Building a dedicated virtual Data Table and enforcing a strict Star Schema was critical for ensuring the YTD and PYTD time-intelligence calculations computed accurately and instantly across millions of rows.